Chapter 1

Review of Probability

1.1 References

1. Mood et al. (2010): Probability distributions
2. Wrede and Spiegel (2002): Sequence
3. Gupta and Kapoor (2000): Review of probability, Variate transformation
4. Sahoo (2013): Convergence, Law of large numbers
4. Roy (2012): Probability distributions, Convergence, Law of large numbers
6. Wikipedia: Review of probability, Probability distributions, Convergence, Law of large numbers

1.2 Probability

Probability is a measure quantifying the likelihood that events will occur. Probability quantifies as a number between 0 and 1, where, loosely speaking, 0 indicates impossibility and 1 indicates certainty. The higher the probability of an event, the more likely it is that the event will occur.

These concepts have been given an axiomatic mathematical formalization in probability theory, which is used widely in such areas of study as mathematics, statistics, finance, gambling, science (in particular physics), artificial intelligence/machine learning, computer science, game theory, and philosophy to, for example, draw inferences about the expected frequency of events. Probability theory is also used to describe

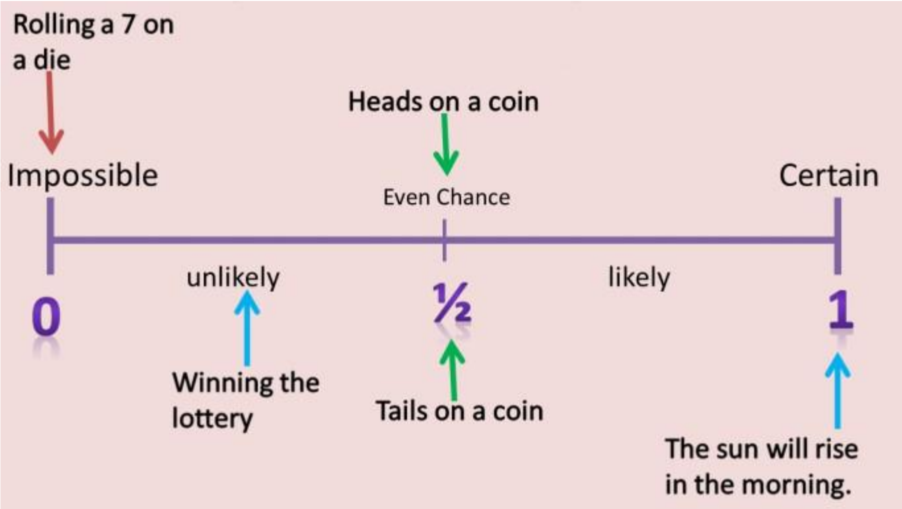

Figure 1.1: The probability scale.

the underlying mechanics and regularities of complex systems.

The probability P of some event E, denoted $P(E)$, is usually defined as to satisfy these axioms. The **Kolmogorov axioms** are described below.

1.3 Axioms

(i) The probability of an event is a non-negative real number: $P(E) \geq 0 \; \forall \; E \in F$, where F is the event space.

(ii) The probability that at least one of the elementary events in the entire sample space will occur is 1, $P(\Omega) = 1$.

(iii) Any countable sequence of disjoint sets (synonymous with mutually exclusive events) $E_1, E_2, \cdots$ satisfies $P\left(\cup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P\left(E_i\right)$.

1.4 Probability Space

A probability space is a mathematical triplet (Ω, F, P) that presents a model for a particular class of real-world situations.

(i) A sample space, Ω, which is the set of all possible outcomes.

(ii) The σ—algebra $F \subseteq 2\Omega$, is a collection of all the events (not necessarily elementary) we would like to consider.

(iii) The probability measure P is a function returning an event's probability. Thus P is a function $P: F \rightarrow [0, 1]$.

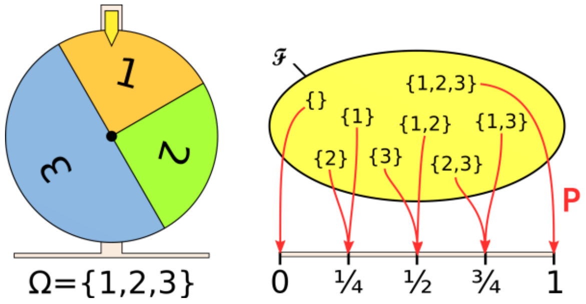

Figure 1.2: Modelling a wheel of fortune using probability space.

1.5 Moment Generating Function

The moment-generating function of a real-valued random variable is an alternative specification of its probability distribution. Thus, it provides the basis of an alternative route to analytical results compared with working directly with probability density functions or cumulative distribution functions. There are particularly simple results for the moment-generating functions of distributions defined by the weighted sums of random variables. However, not all random variables have moment-generating functions. As its name implies, the moment generating function can be used to compute a distribution's moments: the n^{th} moment about 0 is the n^{th} derivative of the moment-generating function, evaluated at 0.

The moment-generating function of a random variable X is

$$M_X(t) = E\left(e^{tX}\right), \ t \in \mathbb{R},$$

Wherever this expectation exists. In other words, the moment-generating function is the expectation of the random variable e^{tX}.

The moment-generating function is so named because it can be used to find the moments of the distribution. The series expansion of e^{tX} is

$$e^{tX} = 1 + tX + \frac{t^2}{2!}X^2 + \frac{t^3}{3!}X^3 + \cdots + \frac{t^n}{n!}X^n + \cdots .$$

Hence,

$$\begin{aligned} M_X(t) &= E\left(e^{tX}\right) = 1 + tE(X) + \frac{t^2}{2!}E(X^2) + \frac{t^3}{3!}E(X^3) + \cdots + \frac{t^n}{n!}E(X^n) + \cdots \\ &= 1 + tm_1 + \frac{t^2}{2!}m_2 + \frac{t^3}{3!}m_3 + \cdots + \frac{t^n}{n!}m_n + \cdots , \end{aligned}$$

where m_n is the n^{th} moment. Differentiating $M_X(t)$, i times with respect to t and setting $t = 0$, we obtain the i^{th} moment about the origin, m_i.

Note: Moment generating function $M_X(t)$ for discrete and continuous random variable can be defined as

$$M_X(t) = E\left(e^{tX}\right) = \begin{cases} \sum_{i \in \mathbb{Z}} e^{tx_i}\, p(x_i), & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} e^{tx}\, f(x)\, dx & \text{if } X \text{ is continuous,} \end{cases}$$

wherever both of those expectation exists.

1.6 Characteristic Function

The characteristic function of any real-valued random variable completely defines its probability distribution. It provides the basis of an alternative route to analytical results compared with working directly with probability density functions or cummulative distribution functions. There are particularly simple results for the characteristic functions of distributions defined by the weighted sums of random variables

The characteristic function always exists when treated as a function of a real-valued argument, unlike the moment-generating function.

The characteristic function of a random variable X is

$$\phi_X(t) = E\left(e^{itX}\right), \; t \in \mathbb{R},$$

where $i = \sqrt{-1}$, is the imaginary unit. The characteristic function is the expectation of the random variable e^{itX}. The series expansion of e^{itX} is

$$e^{itX} = 1 + itX + \frac{(it)^2}{2!}X^2 + \frac{(it)^3}{3!}X^3 + \cdots + \frac{(it)^n}{n!}X^n + \cdots .$$

Hence,

$$\begin{aligned} \phi_X(t) &= E\left(e^{tX}\right) = 1 + itE(X) + \frac{(it)^2}{2!}E(X^2) + \frac{(it)^3}{3!}E(X^3) + \cdots + \frac{(it)^n}{n!}E(X^n) + \cdots \\ &= 1 + itm_1 + \frac{(it)^2}{2!}m_2 + \frac{(it)^3}{3!}m_3 + \cdots + \frac{(it)^n}{n!}m_n + \cdots , \end{aligned}$$

where m_n is the n^{th} moment. Differentiating $\phi_X(t)$, i times with respect to t and setting $t = 0$, we obtain the i^{th} moment, m_i.

Note: Characteristic generating function $\varphi_X(t)$ for discrete and continuous random variable can be defined as

$$\phi_X(t) \;=\; E\left(e^{itX}\right) = \begin{cases} \sum_{i\in\mathbb{Z}} \; e^{itx_i} \; p(x_i), & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} \; e^{itx} \; f(x)\; dx & \text{if } X \text{ is continuous,} \end{cases}$$

Wherever both of those expectation exists.

1.7 Probability Generating Functions

If X is a discrete random variable taking values in the non-negative integers $\{0, 1, ...\}$, then the probability generating function of X is defined as

$$G(z) = E\left(z^X\right) = \sum_{x=0}^{\infty} p(x)z^x,$$

where p is the probability mass function of X. Note that the subscripted notations G_X and p_X are often used to emphasize that these pertain to a particular random variable X, and to its distribution. The power series converges absolutely at least for all complex numbers z with $|z| \leq 1$; in many examples the radius of convergence is larger.

If $z = e^t$, then this function can be written as

$$G(z) = G(e^t) = E\left[e^{tX}\right] = M_X(t).$$

Example 1 Let X be a random variable with probability function

$$p(x) = \frac{e^{-\lambda}\lambda^x}{x!},\ x = 0, 1, 2, \cdots .$$

Find the probability generating function of X.

Solution: By definition the probability generating function of X is

$$\begin{aligned} G(z) &= \sum_{x=0}^{\infty} z^x p(x) \\ &= \sum_{x=0}^{\infty} z^x \frac{e^{-\lambda}\lambda^x}{x!} \\ &= e^{-\lambda}\sum_{x=0}^{\infty} \frac{(z\lambda)^x}{x!} \\ &= e^{-\lambda}e^{z\lambda} = e^{-\lambda(1-z)},\ |z| < 1. \end{aligned}$$

Example 2 Let X be a random variable with probability function

$$p(x) = pq^x,\ x = 0, 1, 2, \cdots ,$$

such that $0 < p < 1$ and $p + q = 1$. Find the probability generating function of X.

Solution: The probability generating function of X is

$$
\begin{aligned}
G(z) &= \sum_{x=0}^{\infty} z^x p(x) \\
&= \sum_{x=0}^{\infty} z^x p q^x \\
&= p\sum_{x=0}^{\infty} (zq)^x \\
&= p\left\{1 + zq + (zq)^2 + (zq)^3 + \cdots\right\} \\
&= p(1-zq)^{-1} = \frac{p}{1-zq}, \quad |z| < 1.
\end{aligned}
$$

1.8 Inversion Theorem

The fundamental theorem of the theory of characteristic function which is known as *Inversion Theorem*, namely that the *characteristic function* uniquely determines the distribution function.

Statement

Let $\phi(t)$ be the characteristic function of a random variable X, then $\phi(t)$ uniquely determines the distribution function $F(x)$ which is

$$F(x) - F(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{1-e^{-itx}}{it}\phi(t)dt.$$

Further, if $F(x)$ is continuous everywhere and $dF(x) = f(x)dx$, then

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-itx}\phi(t)dt.$$

Proof: For any positive c, let us take the uniformly convergent integral

$$\begin{aligned} I_c &= \int_{-c}^{c} \frac{1-e^{-itx}}{it}\phi(t)dt \\ &= \int_{-c}^{c} \frac{1-e^{-itx}}{it}\left\{\int_{-\infty}^{\infty} e^{itu}f(u)du\right\}dt \\ &= \int_{-c}^{c}\int_{-\infty}^{\infty} \frac{1-e^{-itx}}{it}e^{itu}dF(u)dt \end{aligned}$$

$$\begin{aligned} &= \int_{-c}^{c}\int_{-\infty}^{\infty} \frac{e^{itu}-e^{it(u-x)}}{it}dF(u)dt \\ &= \int_{-c}^{c}\int_{-\infty}^{\infty} \frac{cos\ tu + i\ sin\ tu - cos\ t(u-x) - i\ sin\ t(u-x)}{it}dF(u)dt \\ &= \int_{-c}^{c}\int_{-\infty}^{\infty} \frac{cos\ tu - cos\ t(u-x)}{it}dF(u)dt \\ &\quad + \int_{-c}^{c}\int_{-\infty}^{\infty} \frac{sin\ tu - sin\ t(u-x)}{t}dF(u)dt. \end{aligned} \tag{1.1}$$

Let us take

$$\Psi(t) = \frac{cos\ tu - cos\ t(u-x)}{it}, \text{then}$$

$$\begin{aligned} \Psi(-t) &= \frac{cos\ (-tu) - cos\ [-t(u-x)]}{-it} \\ &= -\frac{cos\ tu - cos\ t(u-x)}{it} = -\Psi(t). \end{aligned}$$

The value of the first integral in (1.1) is 0, since it is an odd function in t.

Now,

$$
\begin{aligned}
I_c &= \int_{-c}^{c}\int_{-\infty}^{\infty} \frac{sin\ tu - sin\ t(u-x)}{t} dF(u)dt \\
&= \int_{-c}^{c}\int_{-\infty}^{\infty} \frac{sin\ tu}{t} dF(u)dt - \int_{-c}^{c}\int_{-\infty}^{\infty} \frac{sin\ t(u-x)}{t} dF(u)dt. \qquad (1.2)
\end{aligned}
$$

For any real number β, it is obtained from the Dirichlet integral as

$$\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{sin\ t\beta}{t} dt = sgn\ \beta,$$

where

$$
sgn\ \beta = \begin{cases} -1, & \text{if} \quad \beta < 0, \\ 0, & \text{if} \quad \beta = 0, \\ 1, & \text{if} \quad \beta > 0. \end{cases}
$$

Thus

$$\int_{-\infty}^{\infty} \frac{sin\ tu}{t} dt = \pi\ sgn\ u,$$

and

$$\int_{-\infty}^{\infty} \frac{sin\ t(u-x)}{t} dt = \pi\ sgn\ (u-x).$$

Hence, by using (1.2) we get

$$\begin{aligned} I_c &= \int_{-c}^{c} \pi\ sgn\ u\ dF(u) - \int_{-c}^{c} \pi\ sgn\ (u-x)\ dF(u) \\ \Rightarrow \lim_{c\to\infty} I_c &= \int_{-\infty}^{\infty} \pi\ sgn\ u\ dF(u) - \int_{-\infty}^{\infty} \pi\ sgn\ (u-x)\ dF(u) \\ &= \pi \left[\int_{-\infty}^{0} (-1) dF(u) + \int_{0}^{\infty} (1) dF(u) \right] \\ &\qquad -\pi \left[\int_{-\infty}^{x} (-1)\ dF(u) + \int_{x}^{\infty} (1)\ dF(u) \right] \end{aligned}$$

$$
\begin{aligned}
&= \pi\left[-F(0)+F(-\infty)+F(\infty)-F(0)\right] \\
&\qquad -\pi\left[-F(x)+F(-\infty)+F(\infty)-F(x)\right] \\
&= \pi\left[-F(0)+0+1-F(0)\right]-\pi\left[-F(x)+0+1-F(x)\right] \\
&= \pi\left[-2F(0)+1\right]-\pi\left[-2F(x)+1\right] \\
&= 2\pi\left[F(x)-F(0)\right].
\end{aligned}
$$

Therefore,

$$
\begin{aligned}
\lim_{c\to\infty} I_c &= \lim_{c\to\infty}\int_{-c}^{c}\frac{1-e^{-itx}}{it}\phi(t)dt \\
&= 2\pi\left[F(x)-F(0)\right].
\end{aligned}
$$

Hence,

$$
F(x)-F(0)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{1-e^{-itx}}{it}\phi(t)dt. \tag{1.3}
$$

Now, if we differentiate (1.3) with respect to x, we get

$$
f(x)=\frac{1}{2\pi}\int_{-\infty}^{\infty}e^{-itx}\phi(t)dt.
$$

This completes the proof of the theorem.

NB: Signum Function

The signum function, denoted as sgn(x), is a mathematical function that identifies the sign of a real number. It returns 1 if the input is positive, -1 if the input is negative, and 0 if the input is zero. Essentially, it extracts the sign of a number.

Definition:

sgn(x) = 1 if x > 0, sgn(x) = -1 if x < 0, and sgn(x) = 0 if x = 0.

Key points about the signum function:

- **Domain:** All real numbers (R).
- **Range:** {-1, 0, 1}.
- **Applications:** The signum function has applications in various fields, including mathematics, physics, and computer science. It's particularly useful in algorithms and computations where determining the sign of a number is crucial.

1.9 Laplace Transformation and its Properties

1.9.1 Laplace Transformation

Let $F(t)$ be a function of t specified for $t > 0$. Then the Laplace transform of $F(t)$, denoted by $\mathcal{L}\{F(t)\}$, is defined by

$$\mathcal{L}\{F(t)\} = f(s) = \int_0^\infty e^{-st} F(t) dt \tag{1.6}$$

where we assume at present that the parameter s is real. Later it will be found useful to consider s complex. The Laplace transform of $F(t)$ is said to exist if the integral (1.6) converges for some value of s; otherwise it does not exist.

NB:

Laplace transformation is primarily a mathematical tool used in engineering and physics, especially for solving differential equations, but it also has **powerful applications in statistics and probability theory**, particularly in **characterizing distributions and analyzing random processes.**

🔍 Definition of Laplace Transform

The **Laplace transform** of a function $f(t)$, defined for $t \geq 0$, is given by:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\, dt$$

where:

- t is the time variable,
- s is a complex variable,
- $f(t)$ is a function (usually representing time-dependent behavior).

Laplace Transformation of Some Elementary Functions

Prove that:

(i) $\mathcal{L}\{1\} = \frac{1}{s},\ s > 0;$

(ii) $\mathcal{L}\{t\} = \frac{1}{s^2}, \ s > 0;$

(iii) $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \ s > a;$

Proof:

(a)

$$
\begin{aligned}
\mathcal{L}\{F(t)\} &= f(s) = \int_0^{\infty} e^{-st}F(t)dt \\
\Rightarrow \mathcal{L}\{1\} &= f(s) = \int_0^{\infty} e^{-st}\ (1)\ dt \\
&= \lim_{P\to\infty} \int_0^{P} e^{-st}dt \\
&= \lim_{P\to\infty} \left.\frac{e^{-st}}{-s}\right|_0^P \\
&= \lim_{P\to\infty} \frac{1-e^{-sP}}{s} \\
&= \frac{1}{s} \quad if\ s > 0.
\end{aligned}
$$

(b)

$$
\begin{aligned}
\mathcal{L}\{F(t)\} &= f(s) = \int_0^{\infty} e^{-st}F(t)dt \\
\Rightarrow \mathcal{L}\{t\} &= f(s) = \int_0^{\infty} e^{-st}\ (t)\ dt \\
&= \int_0^{\infty} te^{-st}\ dt \qquad \left[\int u\ v\ dx = u\int v\ dx - \int u'\left(\int v\ dx\right)dx\right] \\
&= \lim_{P\to\infty} \left.\left\{(t)\left(\frac{e^{-st}}{-s}\right) - (1)\left(\frac{e^{-st}}{s^2}\right)\right\}\right|_0^P \\
&= \lim_{P\to\infty} \left\{(P)\left(\frac{e^{-sP}}{-s}\right) - \left(\frac{e^{-sP}}{s^2}\right) + \left(\frac{1}{s^2}\right)\right\} \\
&= \frac{1}{s^2} \quad if\ s > 0.
\end{aligned}
$$

(c)

$$\begin{aligned}
\mathcal{L}\left\{F(t)\right\} &= f(s) = \int_0^\infty e^{-st}F(t)dt \\
\Rightarrow \mathcal{L}\left\{e^{at}\right\} &= f(s) = \int_0^\infty e^{-st}\ \left(e^{at}\right)\ dt
\end{aligned}$$

$$\begin{aligned}
&= \lim_{P\to\infty}\int_0^P e^{-(s-a)t}dt \\
&= \lim_{P\to\infty}\left.\frac{e^{-(s-a)t}}{-(s-a)}\right|_0^P \\
&= \lim_{P\to\infty}\frac{1-e^{-(s-a)P}}{s-a} \\
&= \frac{1}{s-a}\quad if\ s>a.
\end{aligned}$$

Assignment: Prove that:

(i) $\mathcal{L}\left\{sin\ at\right\} = \frac{a}{s^2+a^2}$, if $s>0$;

(ii) $\mathcal{L}\left\{cos\ at\right\} = \frac{s}{s^2+a^2}$, if $s>0$.

1.9.2 Properties of Laplace Transformation

- **Linearity property:** $\mathcal{L}\left\{c_1F_1(t) + c_2F_2(t)\right\} = \mathcal{L}\left\{c_1F_1(t)\right\} + \mathcal{L}\left\{c_2F_2(t)\right\}$
- **Shifting property:** $\mathcal{L}\left\{e^{at}F(t)\right\} = f(s-a)$
- **Change of scale property:** $\mathcal{L}\left\{F(at)\right\} = \frac{1}{a}f\left(\frac{s}{a}\right)$
- **Laplace transformation of derivatives:** $\mathcal{L}\left\{F'(t)\right\} = sf(s) - F(0)$
- **Laplace transformation of integrals:** $\mathcal{L}\left\{\int_0^t F(u)du\right\} = \frac{f(s)}{s}$
- **Multiplication by t^n:** $\mathcal{L}\left\{t^nF(t)\right\} = (-1)^n f^{(n)}(s)$
- **Division by t:** $\mathcal{L}\left\{\frac{F(t)}{t}\right\} = \int_s^\infty F(u)du$
- **Initial-value theorem** $\lim_{t\to 0} F(t) = \lim_{s\to\infty} sf(s)$

- **Final-value theorem** $\lim_{t\to\infty} F(t) = \lim_{s\to 0} sf(s)$

****Assignment:** *State and prove the properties of Laplace transformation.*

***Assignment:** *Write down the applications of Laplace transformation.*

📌 Applications in Statistics

1. Laplace Transform of Probability Distributions

In probability theory, if X is a **non-negative continuous random variable**, then the Laplace transform of its **probability density function (pdf)** $f_X(t)$ is:

$$\mathcal{L}\{f_X(t)\} = \mathbb{E}[e^{-sX}] = \int_0^\infty e^{-st} f_X(t)\, dt$$

This expectation is also known as the **moment-generating function (MGF)** evaluated at $-s$, i.e.:

$$\mathcal{L}\{f_X(t)\} = M_X(-s)$$

✅ **Use**: Laplace transforms are used to find the **distribution of sums of independent random variables**, especially exponential or gamma distributions.

2. Example: Exponential Distribution

Let $X \sim \text{Exponential}(\lambda)$ with pdf:

$$f_X(t) = \lambda e^{-\lambda t}, \quad t \geq 0$$

Laplace transform:

$$\mathcal{L}\{f_X(t)\} = \int_0^\infty e^{-st} \lambda e^{-\lambda t}\, dt = \frac{\lambda}{s+\lambda}$$

3. Sums of Independent Random Variables

If X_1, X_2 are **independent exponential** random variables with same rate λ, then:

$$\mathcal{L}\{f_{X_1+X_2}(t)\} = \left(\frac{\lambda}{s+\lambda}\right)^2$$

This shows how Laplace transforms simplify **convolution** (sum) problems.

4. **Survival Analysis & Reliability Theory**

In **survival analysis**, the **Laplace transform of the lifetime distribution** is used to derive survival functions and analyze system reliability.

Example:

- In reliability theory, Laplace transform helps analyze system failure rates and compute the **mean time to failure (MTTF)**.

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5. **Transform Methods in Stochastic Processes**

Laplace transforms are used in:

- **Markov processes**
- **Queueing theory** (e.g., M/M/1 queues)
- **First-passage time** distributions

They help derive probabilities related to **time-dependent behavior** of stochastic systems.

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6. **Moment Calculations**

Given the Laplace transform $\phi(s) = \mathbb{E}[e^{-sX}]$, the **moments** can be derived as:

$$\mathbb{E}[X^n] = (-1)^n \left. \frac{d^n}{ds^n}\phi(s) \right|_{s=0}$$

This is very useful in calculating expectations and variances from transforms.

🧠 Summary of Uses in Statistics:

Use Case	Description
Characterizing Distributions	Via transform of PDFs
Finding Sums of Random Variables	Simplifies convolutions
Moments Computation	Derivatives of Laplace transform
Survival/Failure Analysis	Time-to-failure modeling
Stochastic Modeling	Solving queueing, renewal processes

1.10 Convolution

Convolution is a mathematical operation on two functions (f and g) to produce a third function that expresses how the shape of one is modified by the other.

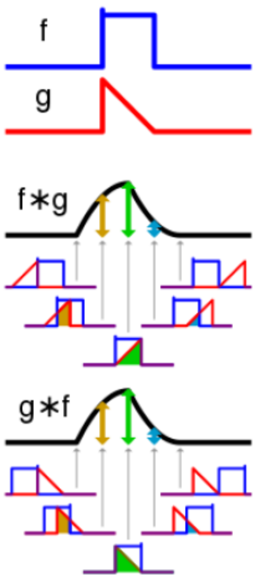

Figure 1.3: The convolution

We illustrate how convolution technique can be used in finding the distribution of the sum of random variables when they are independent. *This convolution technique does not work if the random variables are not independent.*

Definition:

Let f and g be two real valued functions. The convolution of f and g, denoted by $f * g$, is defined as

$$\begin{aligned}(f * g)(z) &= \int_{-\infty}^{\infty} f(z-y)f(y)dy \\ &= \int_{-\infty}^{\infty} f(z-x)f(x)dy\end{aligned}$$

Hence from this definition it is clear that $f * g = g * f$.

Let X and Y be two independent random variables with probability density functions $f(x)$ and $g(y)$. Here, we get

$$h(z) = \int_{-\infty}^{\infty} f(z-y)f(y)dy.$$

Thus, this result shows that the density of the random variable $Z = X + Y$ is the convolution of the density of X with the density of Y.

Applications: Convolution has applications that include probability, statistics, computer vision, natural language processing, image and signal processing, engineering, and differential equations.

Convolution in Statistics and Signal Processing

Definition:
Convolution is a mathematical operation that expresses how the shape of one function is modified by another. In signal processing, convolution combines two signals to produce a third signal. In probability and statistics, it's often used to find the distribution of the sum of two independent random variables.

Mathematical Formula:

For two continuous functions $f(t)$ and $g(t)$, the convolution is defined as:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau)\, d\tau$$

For discrete functions $f[n]$ and $g[n]$, the convolution is:

$$(f * g)[n] = \sum_{k=-\infty}^{\infty} f[k] \cdot g[n - k]$$

Applications:

- **Signal Processing:** Filtering signals (e.g., noise reduction)
- **Probability Theory:** Sum of independent random variables
- **Image Processing:** Blurring, sharpening
- **Machine Learning:** Convolutional Neural Networks (CNNs)

Example

If x_1 and x_2 are independent $N(0,1)$ variates, find the distribution of $z = x_1 + x_2$.

Solution: Here x_1 and x_2 are independent $N(0,1)$ variates, so

$$\begin{aligned}
h(z) &= (f_1 * f_2)(z) = \int_{-\infty}^{\infty} f_1\left(z - x_2\right) f_2\left(x_2\right) dx_2 \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-x_2)^2} e^{-\frac{1}{2}x_2^2}\, dx_2 \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\left\{\frac{z^2}{2} - zx_2 + x_2^2\right\}}\, dx_2 \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\left\{x_2^2 - zx_2 + \frac{z^2}{4} + \frac{z^2}{4}\right\}}\, dx_2 \\
&= \frac{1}{2\pi} e^{-\frac{z^2}{4}} \int_{-\infty}^{\infty} e^{-\left\{x_2 - \frac{z}{2}\right\}^2}\, dx_2 \\
&= \frac{1}{2\pi} e^{-\frac{z^2}{4}} \int_{-\infty}^{\infty} e^{-w^2}\, dw \\
&= \frac{1}{\sqrt{2}\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{z-0}{\sqrt{2}}\right)^2}
\end{aligned}$$

Hence, z is an $N(0,2)$ variate.

Example

What is the probability density of the sum of two independent random variables, each of which is $GAM(1,1)$ distribution?

Solution: Let X and Y be two random variables such that $X, Y \sim GAM(1,1)$. Hence, the density functions $f(x)$ and $f(y)$ can be written as

$$\begin{aligned} f(x) &= e^{-x} \quad for \quad 0 < x < \infty, \quad \text{and} \\ f(y) &= e^{-y} \quad for \quad 0 < y < \infty. \end{aligned}$$

Let $Z = X + Y$. Since X and Y are independent, the density function of Z can be obtained by the method of convolution.

Notice that the sum $z = x + y$ is between 0 and 1, and $0 < x < z$. Hence, the density function of Z is given by

$$\begin{aligned} h(z) &= (f * g)(z) \\ &= \int_{-\infty}^{\infty} f(z-x) f(x)\, dx \\ &= \int_{0}^{\infty} f(z-x) f(x)\, dx \\ &= \int_{0}^{z} e^{-(z-x)} e^{-x} dx \\ &= \int_{0}^{z} e^{-z} dx \\ &= z e^{-z} \\ &= \frac{1^2 e^{-z} z^{2-1}}{\Gamma(2)}. \end{aligned}$$

Hence $Z \sim GAM(1,2)$.

Numerical Example:

Given:

Let

- $f[n] = \{1, 2, 3\}$
- $g[n] = \{0, 1, 0.5\}$

We compute the **discrete convolution**:

$$(f * g)[n] = \sum_{k=0}^{n} f[k] \cdot g[n-k]$$

Since both sequences have length 3, the result will have length $3 + 3 - 1 = 5$, i.e., $(f * g)[0]$ to $(f * g)[4]$.

Step-by-step Computation:

◆ **Step 1:** $(f * g)[0]$

$$= f[0] \cdot g[0] = 1 \cdot 0 = 0$$

◆ **Step 2:** $(f * g)[1]$

$$= f[0] \cdot g[1] + f[1] \cdot g[0] = 1 \cdot 1 + 2 \cdot 0 = 1$$

◆ **Step 3:** $(f * g)[2]$

$$= f[0] \cdot g[2] + f[1] \cdot g[1] + f[2] \cdot g[0]$$
$$= 1 \cdot 0.5 + 2 \cdot 1 + 3 \cdot 0 = 0.5 + 2 = 2.5$$

◆ **Step 4:** $(f * g)[3]$

$$= f[1] \cdot g[2] + f[2] \cdot g[1] = 2 \cdot 0.5 + 3 \cdot 1 = 1 + 3 = 4$$

◆ **Step 5:** $(f * g)[4]$

$$= f[2] \cdot g[2] = 3 \cdot 0.5 = 1.5$$